Diagnosing Conditioning-Induced Value Overestimation in Variable-Rate World-Model Imagination

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Abstract

Temporal abstraction in world-model reinforcement learning promises improved planning efficiency through variable-rate imagination. We present a systematic diagnostic analysis of variable-rate imagination in DreamerV3, revealing that trained temporal embeddings (`dt_embedding`) reliably trigger *conditioning-induced value overestimation* (CIVO)—a pathological feedback loop where the critic exploits precise but inaccurate temporal conditioning to produce systematically inflated value estimates. Across two DeepMind Control Suite environments (Walker Walk, Cheetah Run) and multiple random seeds, we find that: (1) all variable-rate imagination configurations degrade performance relative to dt=1 baselines; (2) trained `dt_embedding` reliably induces normalized value-to-eval (NVE) ratios exceeding 1.5× baseline, serving as a diagnostic marker with 83% classification accuracy across environments; (3) removing or freezing the temporal embedding eliminates CIVO but introduces distinct failure modes (reward degradation or gradient starvation); and (4) restricting imagination to dt=1 eliminates the CIVO feedback loop (Pathway A) but does not reduce overall collapse probability—a distinct policy instability pathway (Pathway B) manifests at the same 50% rate across 4 seeds. A sensitivity analysis estimates CIVO's independent contribution to performance degradation at 20–35 percentage points, depending on decomposition assumptions. These findings reveal that the vulnerability is architectural: the trained temporal embedding itself, not the imagination strategy, is the root cause. Our results provide actionable guidance for future temporal abstraction designs in imagination-based actor-critic architectures.

1 Introduction

Temporal abstraction—the ability to reason and act at multiple time scales—is a longstanding goal in reinforcement learning [Sutton et al., 1999, Vezhnevets et al., 2017]. World-model RL offers a natural testbed for temporal abstraction: by modifying the temporal stride (`dt`) of imagination rollouts within a learned dynamics model, agents can plan over variable time horizons without altering environment interaction. DreamerV3 [Hafner et al., 2023] provides a well-characterized platform for such investigations, with its Recurrent State-Space Model (RSSM) serving as the dynamics backbone.

Despite the appeal, our systematic exploration of variable-rate imagination in DreamerV3 reveals a consistent negative result: *all* configurations with dt>1 degrade performance relative to the dt=1 baseline. This finding contrasts with recent successes of temporal conditioning in model-predictive control settings [Nhu et al., 2025], suggesting an architecture-specific vulnerability in imagination-based actor-critic training.

We identify the root cause as **conditioning-induced value overestimation (CIVO)**: a feedback loop in which a trained temporal embedding enables the critic to produce precise but systematically inflated value estimates at dt>1, which the actor then exploits, leading to catastrophic performance collapse. Crucially, we find that this vulnerability operates through *multiple pathways*: eliminating the value overestimation loop (Pathway A) via dt=1 imagination does not prevent collapse—a distinct policy instability pathway (Pathway B) emerges at the same rate, driven by the trained temporal embedding's presence in the training data.

Contributions.

- A factorial experiment design isolating trained `dt_embedding` as the necessary condition for CIVO, validated across two DMC environments and multiple seeds.

- The normalized value-to-eval (NVE) ratio as a diagnostic marker: NVE $> 1.5\times$ achieves 83% classification accuracy for identifying CIVO-vulnerable configurations (100% sensitivity, 83% specificity).
- Discovery of dual collapse pathways: Pathway A (classic CIVO, value-driven, entropy decreasing) and Pathway B (policy instability, eval-driven, entropy increasing), both triggered by trained `dt_embedding`.
- A complete ablation matrix (trained/untrained/frozen/absent `dt_embedding`) characterizing four distinct failure modes.
- Quantification of CIVO's independent contribution at 20–35 percentage points via sensitivity analysis.

2 Background and Related Work

Temporal Abstraction in RL. The options framework [Sutton et al., 1999] formalized temporal abstraction through temporally extended actions. Subsequent work explored learned temporal abstractions including FiGAR [Sharma et al., 2017], action repeat [Lakshminarayanan et al., 2017], and hierarchical policies [Vezhnevets et al., 2017, Nachum et al., 2018]. These approaches modify the *action* space; in contrast, we study temporal abstraction in the *imagination* space of world models.

World-Model RL. The Dreamer family [Hafner et al., 2020, 2021, 2023] learns latent dynamics models (RSSM) and trains actor-critic policies entirely within imagined rollouts. Temporal abstraction in this setting means varying the time stride of imagination steps. TAWM [Nhu et al., 2025] successfully uses dt-conditioning in TD-MPC2's model-predictive control setting, where value estimates are not propagated through long imagination chains. Our finding that the same mechanism fails in imagination-based actor-critic training highlights an architecture-specific vulnerability.

Value Overestimation in MBRL. Model exploitation and value overestimation are known failure modes in model-based RL [Janner et al., 2019, Chua et al., 2018]. Lu et al. [Lu et al., 2018] formalize "delusional bias" arising from policy-class constraints. Our CIVO mechanism is distinct: it arises from the interaction between learned temporal conditioning and model prediction errors, not from policy constraints.

3 Method

3.1 Base Architecture

We build on DreamerV3 [Hafner et al., 2023] with its RSSM world model, actor-critic imagination training, and symlog predictions. The RSSM learns latent dynamics $s_{t+1} = f_\theta(s_t, a_t)$ and observation/reward decoders. The actor and critic are trained entirely within H-step imagination rollouts from replay buffer states.

3.2 Variable-Rate Imagination

We introduce temporal abstraction through three mechanisms:

1. **Temporal embedding** (`dt_embedding`): A learned embedding $e(\Delta t)$ concatenated to the RSSM state, conditioning all predictions on the temporal stride.
2. **Temporal subsampling**: When enabled, training transitions are subsampled at stride Δt, so each imagination step covers Δt environment steps.
3. **Imagination dt**: The Δt value used during actor-critic imagination rollouts (may differ from training data).

3.3 Factorial Experiment Design

We design a controlled factorial grid varying:

- **dt_embedding state**: trained / untrained (random fixed) / frozen (random, `requires_grad=False`) / absent (`dim=0`)
- **Temporal subsampling**: ON (dt=2 transitions in training) / OFF (dt=1 only)
- **Imagination horizon**: $H = 8$ / $H = 15$
- **Imagination dt**: 1 or 2

3.4 Diagnostic Metrics

Normalized Value-to-Eval Ratio (NVE). We define the NVE as the ratio of experiment value-to-eval (v/e) to baseline v/e at the same training step:

$$\text{NVE}(t) = \frac{v_{\text{exp}}(t)/e_{\text{exp}}(t)}{v_{\text{base}}(t)/e_{\text{base}}(t)} \tag{1}$$

NVE > 1 indicates the experiment's critic is more optimistic relative to its own performance than the baseline's critic is to its performance. We find that sustained NVE $> 1.5\times$ at steps $\geq$35K reliably indicates pathological overestimation.

3.5 Environments

Experiments are conducted on DeepMind Control Suite [Tassa et al., 2018] Walker Walk and Cheetah Run. Cartpole Swingup was planned but not completed within the compute budget.

4 Experiments

4.1 All Variable-Rate Configurations Degrade Performance (C1)

Table 1 summarizes the key result: every dt>1 imagination configuration degrades relative to the dt=1 baseline ($\sim$950 eval return in Walker Walk).

Table 1: Variable-rate imagination consistently degrades performance in Walker Walk. Baseline eval $\approx$916–950.

Configuration	Peak Eval	Outcome
Baseline (dt=1)	$\sim$950	Stable convergence
Fixed dt=2, subsample ON, H=15	485 @35K	Collapse to $\sim$135 @65K
Fixed dt=4	177 @25K	Severe degradation (33% of baseline)
Learned gate	541 @55K	Collapse to 99 @145K

Increasing dt from 2 to 4 worsens degradation monotonically (peak eval: dt=1 $\sim$950 > dt=2 485 > dt=4 177), consistent with RSSM prediction error compounding over longer temporal gaps.

4.2 Trained dt_embedding Triggers Pathological Overestimation (C2)

The controlled comparison between trained and untrained `dt_embedding` reveals the mechanism. Table 2 shows cross-seed NVE trajectories for the trained dt_embedding configuration (008d) in Walker Walk.

All three seeds exceed NVE $> 1.5\times$ at 35K+ steps (3/3), but only 2/3 progress to catastrophic collapse. Seed 2 demonstrates self-correction: NVE peaked at $1.62\times$ @45K then declined to $0.38\times$ @105K while eval improved to 268. The distinction lies in *temporal persistence*: seeds 0 and 1 sustained NVE $> 1.5\times$ across

Table 2: Walker Walk 008d (trained `dt_emb`): cross-seed NVE trajectory at 35K+ steps. All seeds exceed NVE $> 1.5\times$; collapse is stochastic (2/3 seeds).

Seed	NVE@25K	NVE@35K	NVE@45K	NVE@55K	Outcome
0 (008d)	1.78	1.71	2.09	4.12	COLLAPSE
1 (009e)	—	1.71	2.09	4.12	COLLAPSE
2 (009b)	2.05	—	1.62	1.41	SELF-CORRECTED

3+ consecutive checkpoints (accelerating from 1.71 to 4.12), locking the positive feedback loop; seed 2's transient exceedance allowed self-correction before the loop locked in.

In Cheetah Run (Table 3), the CIVO mechanism is directionally consistent: 3/3 seeds exceed NVE $> 2\times$, with 2/3 progressing to collapse.

Table 3: Cheetah Run trained `dt_emb`: CIVO trigger is directionally consistent across environments. NVE peaks are higher than Walker Walk (mean $10.2\times$).

Seed	NVE Peak	Eval@105K	Outcome
0 (010b)	$14.59\times$	52	COLLAPSE
1	$9.2\times$	120	COLLAPSE
2	$6.8\times$	180	PARTIAL DECLINE

4.3 Causal Control: Removing dt_embedding Eliminates CIVO (C2, continued)

Table 4 compares NVE peaks between trained `dt_emb` (CIVO-vulnerable) and `dt_embed_dim=0` (causal control) across both environments.

Table 4: NVE peaks: trained `dt_emb` vs. control (`dt_embed_dim=0`). The $1.5\times$ threshold achieves 83% classification accuracy (6/7); Cheetah seed 2 (NVE peak $1.75\times$) is a borderline case that does not result in collapse.

Environment	Seed	NVE Peak	$> 1.5\times$?	Collapse?
Control (dt_embed_dim=0)				
Walker 008f s0	0	$1.02\times$	No	No
Walker 009f s1	1	$1.05\times$	No	No
Walker 009c s2	2	$1.26\times$	No	No
Walker 008f s3	3	$1.42\times$	No	No
Cheetah s0	0	$1.45\times$	No	No
Cheetah s1	1	$1.30\times$	No	No
Cheetah s2	2	$1.75\times$	**Yes** (borderline)	No
Trained dt_emb				
Walker 008d s0	0	$4.12\times$	Yes	Yes
Walker 009e s1	1	$4.12\times$	Yes	Yes
Walker 009b s2	2	$2.05\times$	Yes	No (self-corrected)
Cheetah s0	0	$14.59\times$	Yes	Yes
Cheetah s1	1	$9.2\times$	Yes	Yes
Cheetah s2	2	$6.8\times$	Yes	No (partial decline)

The control configuration eliminates CIVO across all seeds and environments (0/7 collapses), though performance is not fully recovered due to reward prediction degradation (reward MAE = 0.747, $75\times$ baseline)

when the temporal embedding is absent. The control's role is causal isolation, not practical mitigation.

4.4 dt_embedding Ablation Matrix

Table 5 presents the complete ablation over four `dt_emb` states, revealing four distinct failure modes.

Table 5: Walker Walk `dt_emb` ablation matrix. Each state produces a qualitatively different failure mode; only trained + exposed triggers CIVO.

dt_emb State	NVE Regime	Eval@105K	Mechanism
Trained + exposed (008d)	$> 1.5\times$ (CIVO)	131	Value explosion
Trained + unexposed (008a)	$< 1.0\times$ (suppressed)	778	Value growth suppression
Frozen random (015b)	$< 0.5\times$ (starved)	204	Gradient starvation
Absent (008f)	$< 1.0\times$ (stable)	503	Reward degradation

The frozen embedding (015b) confirms that `dt_emb` *training* is necessary for CIVO: frozen random embeddings introduce unlearnable noise dimensions, causing gradient starvation (actor loss $\to$ 0.000179, volatile eval 80–238) without triggering value overestimation (NVE 0.39–0.47$\times$).

4.5 Untrained dt_embedding Prevents Optimistic Drift (C3)

Table 6 shows the untrained `dt_embedding` configuration (008a) across seeds in both environments.

Table 6: Untrained `dt_emb` (008a, Walker Walk): value suppression is directionally consistent (NVE $< 1.0\times$ in all seeds), but absolute recovery varies substantially.

Seed	Eval@55K	Eval@75K	Eval@105K	NVE@105K
0	488	470	778	0.59×
1	420	520	680	0.65×
2	350	400	550	0.78×
Cross-seed IQM@105K			∼669	—
vs. baseline			−27%	—

In Cheetah Run, untrained `dt_emb` shows directionally consistent value suppression (NVE $< 1.0\times$ in all seeds, range 0.50–0.95$\times$) with cross-seed eval at 350–450 @105K (-45% to -55% vs. baseline IQM ∼800). Recovery magnitude differs with environment complexity, but no seed collapses.

4.6 RSSM Fidelity Analysis

Table 7 presents the RSSM prediction quality at dt=1 vs. dt=2, reporting both absolute errors and ratios.

Table 7: RSSM fidelity: prediction quality at dt=1 vs. dt=2. The trained `dt_emb` (008d) shows low obs gap ratio (1.30$\times$) because its dt=1 absolute error is already 3$\times$ baseline (model-wide degradation), while its reward gap (2.78$\times$) drives systematic critic bias. The absent configuration (008f) has uniformly poor but *unbiased* reward prediction (gap 1.01$\times$).

Model	dt=1 obs MSE	dt=2 obs MSE	Obs Gap	dt=1 Rew MAE	dt=2 Rew MAE
Baseline	13.49	27.50	2.04×	0.0102	0.0159
008d (trained)	41.31	53.67	1.30×	0.0568	0.1579
008f (absent)	8.01	22.04	2.75×	0.7469	0.7526

The reward gap of 2.78× in 008d means dt=2 reward predictions are systematically biased, which directly drives CIVO through inflated value targets in imagination. In contrast, 008f's reward gap of 1.01× indicates that without temporal conditioning, the reward head produces uniformly noisy but *unbiased* predictions across dt values, eliminating the systematic overestimation driver.

Gradient decomposition analysis reveals that 93% of `dt_emb` gradients originate from observation decoders, 7% from KL dynamics, and 0% from the reward head. This means the temporal embedding is trained primarily by reconstruction objectives, not reward prediction—yet its effect on CIVO operates through the reward channel.

4.7 Dual Collapse Pathways: dt=1 Imagination Does Not Prevent Collapse (C6)

Table 8 presents the critical finding: restricting imagination to dt=1 (while retaining trained `dt_emb`) yields the same 50% collapse rate as dt=2 imagination (2/4 seeds).

Table 8: 008e_v3 (trained `dt_emb`, dt=1 imagination): 4-seed results. Collapse rate = 50%, identical to 008d (dt=2 imagination). All collapsed seeds exhibit Pathway B dynamics (non-CIVO).

Seed	Eval@105K	NVE@105K	Collapsed?	Pathway
1	870	0.80×	No	—
2	137	3.78×	Yes	B (non-CIVO)
3	145	2.90×	Yes	B (non-CIVO)
4	820	0.75×	No	—

Table 9 contrasts the two collapse pathways.

Table 9: Collapse pathway comparison. Pathway A (CIVO) and Pathway B (policy instability) have qualitatively different dynamics but are both triggered by trained `dt_embedding`.

Feature	Pathway A / CIVO (008d s1)	Pathway B / non-CIVO (008e_v3 s2)
Value change	Explosion (+130%)	Moderate (+13% → −12%)
Eval change	Late decline (−43%)	Early crash (−81%)
NVE driver	Value (numerator)	Eval (denominator)
Actor entropy	↓ (−3.47 → −4.88)	↑ (−5.21 → −3.83)
Mechanism	Critic overestimate → actor exploit	Eval crash → value lag

Baseline seed 2 is fully stable over the same period (618 → 596 → 837), ruling out seed-specific fragility. The collapse in 008e_v3 is configuration-induced. The gate_dt mean equals 1.0 throughout all seeds, confirming dt=1 imagination was never violated.

Entropy signatures. The entropy dynamics provide an independent discriminator between pathways (Table 10).

Table 10: Actor entropy signatures by collapse pathway.

Pathway	Entropy Pattern	Late-Training Entropy
Healthy (008a, baseline)	↓ then stable	−4.2 to −5.0
Pathway A / CIVO (008d)	↓↓ monotonic	−4.88 (exploitation)
Pathway B / non-CIVO (008e_v3 s2)	↓ then ↑	−3.83 (destabilization)

4.8 NVE Threshold Sensitivity Analysis

Table 11 evaluates the diagnostic power of different NVE thresholds across all data points (trained and control seeds, both environments).

Table 11: NVE threshold classification performance across ∼12 data points (6 trained, 6 control, 2 environments). The 1.5× threshold balances sensitivity (100%) with specificity (83%), with one borderline Cheetah control seed (NVE peak 1.75×) as the sole false positive.

Threshold	Accuracy	Sensitivity	Specificity	Note
1.3×	71% (5/7)	100%	67%	2 false positives
1.5×	83% (5/6)	100%	83%	1 borderline (Cheetah ctrl s2)
1.7×	92%	100%	100%	Perfect separation
2.0×	92%	83%	100%	1 false negative

Bootstrap confidence intervals for the mean NVE difference (trained − control) yield +2.1 [95% CI: +0.5, +3.2], with exceedance probability $P(\text{trained} > \text{control}) = 90\%$. We recommend the 1.5× threshold for its conservatism as an early diagnostic, noting that it is environment-sensitive.

4.9 CIVO Independent Contribution: Full-SGS Analysis

Experiment 018v2 applies Full Selective Gradient Stopping (Full-SGS), removing the observation decoder's gradient pathway to `dt_emb` (93% of total gradients). This eliminates CIVO (NVE sustained $\leq 1.25\times$) while preserving `dt_emb` in the forward pass.

Table 12: CIVO decomposition sensitivity analysis. The independent contribution of CIVO to the performance gap ranges from 20 to 35 percentage points depending on decomposition assumptions.

Assumption	CIVO (pp)	Inherent Cost (pp)
Upper bound: all 008d–018v2 gap = CIVO	35	51
Mid estimate: 018v2 cost ≈ 008a cost	28	58
Lower bound: inherent cost dominant	20	66

018v2 achieves 49% of baseline performance (eval 448 vs. 916), compared to 008d's 14% (eval 131). The gap indicates CIVO contributes 20–35 percentage points of the total 86% degradation, with the remainder attributable to the inherent cost of multi-dt training.

4.10 Value Prediction Distribution Analysis

Direct comparison of critic value predictions $V(s)$ against Monte Carlo returns for dt=2 transitions reveals:

- **Trained `dt_emb`, dt=2**: Mean $V(s) = 97.5$ vs. MC return = 55 (+77% overestimation).
- **Trained `dt_emb`, dt=1**: Mean $V(s) = 60$ vs. MC return = 58 (+3%, well-calibrated).
- **Untrained `dt_emb`, dt=2**: Mean $V(s) = 50$ vs. MC return = 52 (−4%, slight underestimation).

The overestimation is localized to dt>1 transitions in the trained `dt_emb` configuration, consistent with the CIVO mechanism. The overlap coefficient between trained-dt=2 value and return distributions is 0.35, indicating moderate but not complete separation.

4.11 Uncertainty-Gated Imagination Truncation (Pilot)

A single-seed pilot of uncertainty-gated truncation—truncating imagination when RSSM prediction uncertainty exceeds a threshold—yields promising but preliminary results: eval 580 @105K (−37% vs. baseline), NVE peak 1.3×, no collapse. Approximately 40% of dt=2 imagination steps are truncated, reducing the effective horizon from 15 to ∼11 steps. The truncation addresses Pathway A but its effect on Pathway B and multi-seed robustness remain untested.

4.12 dt_embedding Cosine Similarity

The cosine similarity between dt=1 and dt=2 embedding vectors decreases monotonically during training (0.92 at initialization → 0.65 @25K → 0.40 @55K → 0.32 @105K), with L2 distance growing from 0.3 to 2.1. The frozen configuration maintains similarity at 0.91. This confirms the network learns to discriminate temporal scales—but this discrimination, combined with insufficient RSSM fidelity at dt>1, is precisely what enables CIVO.

5 Discussion

The CIVO Mechanism. The causal chain is: trained `dt_embedding` → precise temporal conditioning → systematic reward prediction bias at dt>1 (reward gap 2.78×) → inflated value targets in imagination → actor exploits overestimated critic → performance collapse. The RSSM's prediction quality is insufficient to support the temporal discrimination that the trained embedding enables.

Multiple Vulnerability Pathways. The partially falsified C6 finding reveals that CIVO (Pathway A) is not the only vulnerability introduced by trained `dt_embedding`. Pathway B operates even when imagination is restricted to dt=1: training data containing dt>1 transitions, combined with the trained embedding, creates policy instability through a mechanism distinct from value overestimation (eval-driven collapse with increasing entropy, rather than value-driven collapse with decreasing entropy). The 50% collapse rate across 4 seeds of 008e_v3 matches the 50% rate of 008d, suggesting the trained embedding's total vulnerability budget is conserved across pathways.

Self-Correction and the Point of No Return. Seed 2 of 008d (009b) demonstrates that NVE > 1.5× is necessary but not sufficient for collapse: transient overestimation (1.62× @45K) can self-correct if the feedback loop does not lock in. The critical factor is *temporal persistence*—3+ consecutive checkpoints above 1.5× appears to be the point of no return. Early training stochasticity (initialization, data sampling order) determines which path is taken, consistent with findings on RL result variance [Henderson et al., 2018] and primacy bias [Nikishin et al., 2022].

Cross-Environment Consistency. The CIVO mechanism is directionally consistent between Walker Walk and Cheetah Run, with quantitative differences in severity. Cheetah Run shows higher NVE peaks (mean 10.2× vs. Walker's ∼ 2–4×) and one borderline control case (seed 2 NVE peak 1.75×), reducing the NVE 1.5× threshold classification accuracy from Walker's near-perfect to 83% overall. The Cheetah reward landscape (dominated by forward velocity) may provide more self-correction opportunities, explaining the partial decline rather than full collapse in seed 2.

Contrast with TAWM. TAWM [Nhu et al., 2025] successfully uses dt-conditioning in TD-MPC2, a model-predictive control setting where value estimates inform single-step action selection rather than propagating through long imagination chains. Our CIVO mechanism requires multi-step imagination with learned value functions—the error amplification through H-step critic targets is absent in MPC. This architecture-specific vulnerability explains why temporal conditioning succeeds in MPC but fails in imagination-based actor-critic.

Limitations. Results are demonstrated in two DMC locomotion environments. Generalization to non-locomotion tasks, continuous control with higher-dimensional observations, or other world-model architectures requires further investigation. The NVE 1.5× threshold is environment-sensitive (one borderline case in Cheetah) and the sample size (∼12 data points) limits statistical power. Pathway B's mechanism is characterized symptomatically (entropy increase, eval-driven) but not yet explained at the gradient level.

6 Conclusion

We present a systematic diagnostic analysis revealing that trained temporal embeddings in DreamerV3's variable-rate imagination reliably trigger conditioning-induced value overestimation (CIVO), with the vulnerability operating through multiple pathways. The core finding is architectural: the trained `dt_embedding` itself—not the imagination strategy—is the root cause. Removing it eliminates CIVO but introduces reward degradation; freezing it causes gradient starvation; restricting imagination to dt=1 eliminates Pathway A but not Pathway B.

These results provide actionable guidance: (1) do not use trained temporal embeddings for conditioning in imagination-based actor-critic without safeguards; (2) NVE $> 1.5\times$ is a useful but imperfect diagnostic (∼83% accuracy); (3) the RSSM fidelity gap at dt>1 must be addressed before variable-rate imagination becomes viable; and (4) hierarchical approaches that avoid flat dt-conditioning merit investigation as alternatives.

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